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Small Target Detection in UAV Remote Sensing Images Using Deep Learning

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***Abstract*—Along with the rapid development of unmanned aerial vehicle (UAV) technology, UAV remote sensing images have been widely used in extensive fields. Nevertheless, the detection of small targets in images inevitably faces numerous severe challenges. Against this background, this paper deeply explores the real-time small target detection method by UAV remote sensing images based on deep learning, explaining the relevant algorithms and architecture in detail. Based on YOLOv8, this paper proposes a small object detection model, which incorporates a feature enhancement, named FPE-CAG-YOLO. Meanwhile, this paper validates the effectiveness of the proposed method through experiments to significantly improve the accuracy and real-time capability of small target detection. Extensive experiments conducted on the VisDrone dataset, which is UAV dataset. The FPE-CAG-YOLO improves the baseline by 8.6%(mAP50).**

Keywords—deep learning, unmanned aerial vehicle remote sensing image, small target detection

I. Introduction

With a series of advantages such as flexibility, high efficiency, and low cost, unmanned aerial vehicle (UAV) remote sensing has been widely used in agriculture, the environment, urban planning, and other fields [1]. Despite the rich image information obtained, the detection of small targets has consistently been a difficult problem. Given the small proportion of pixels and the unapparent features of these small targets, the traditional detection methods failed to achieve significant accuracy and real-time performance. From a theoretical standpoint, research on small-target detection in UAV remote sensing images can enrich the theoretical framework for feature extraction and fusion in deep learning under "low-resolution, high-interference" conditions, offering novel technical approaches for multi-scale target detection. From an application perspective, efficient small-target detection technology can substantially enhance the utilization efficiency of UAV remote sensing data: in agriculture, it enables early and precise identification of pests and diseases, thereby reducing pesticide use; in urban traffic management, it facilitates real-time monitoring of small vehicle violations, improving traffic control accuracy; in emergency rescue operations, it swiftly locates small trapped targets in disaster-stricken areas, providing crucial support for rescue decision-making.

At present, deep learning-based object detection algorithms are mainly divided into two categories: two-stage algorithms and single-stage algorithms. The two-stage algorithm, exemplified by Faster R-CNN [2], generates candidate regions via a Region Proposal Network (RPN) and then performs classification and regression. It boasts high detection accuracy but is computationally intensive, posing challenges in meeting real-time demands. To enhance small object detection, researchers have improved Faster R-CNN, such as the Feature Pyramid Network (FPN) structure proposed by Lin et al. [3], which boosts small object detection capability by fusing multi-scale features. Nevertheless, it still encounters the problem of losing semantic information in deep-level features. The single-stage algorithm, with the YOLO series as its core [4], adopts an end-to-end detection framework, showcasing remarkable real-time performance advantages. YOLOv5 improves feature extraction efficiency by utilizing Mosaic data augmentation and the CSPDarknet backbone network. YOLOv8 [5] further integrates FPN and PANet [6] architectures to optimize multi-scale feature fusion. Nevertheless, neither model has developed a specialized feature pyramid tailored to the small target characteristics inherent in UAV remote sensing images, resulting in inadequate feature extraction for small targets. Moreover, attention mechanisms have found widespread application in small target detection, exemplified by the SE module and CBAM module [7], which enhance detection accuracy by emphasizing critical feature channels or spatial information. However, current research predominantly employs a single attention mechanism in isolation, neglecting the necessity to mitigate multi-dimensional interference in the complex backgrounds of UAV images.

Current object detection algorithms exhibit three significant drawbacks when applied to small target detection in UAV remote sensing images: First, traditional feature pyramids (e.g., FPN and PANet), which acquire multi-scale features through successive downsampling and upsampling, are prone to losing detailed features of small targets. Second, a single attention mechanism finds it challenging to simultaneously mitigate both channel and spatial interference in complex backgrounds, causing small target features to be overshadowed. Third, while some improved algorithms have boosted detection accuracy, they have also introduced substantial computational overhead, rendering them unsuitable for real-time detection scenarios in drone applications. To address these issues, this paper aims to develop a small target detection model that achieves a balance between accuracy and real-time performance thereby resolving the aforementioned technical challenges.

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II. Relevant Theory Foundations

A. Deep Learning

Conceptually, deep learning is defined as a kind of machine learning technology based on an artificial neural network, which learns a large amount of data by constructing a neural network model with multiple hidden layers, thereby automatically extracting data-related features. Within remote sensing, CNNs are the most widely applied deep learning models for land cover mapping and object detection [8]. Through various components such as the convolutional layer, pooling layer, and fully connected layer, CNN is capable of extracting local and global features of images efficiently. CNN, for example, can extract various features of the image, such as edges and textures, through convolution operation for the convolution kernel in the convolutional layer that slides on the image.

B. Features of UAV Remote Sensing Images

UAV remote sensing images demonstrate a series of remarkable characteristics, such as high resolution and flexible viewing angles. Nevertheless, they exposed such defects as small target sizes and complex backgrounds. In comparison with satellite remote sensing images, the images obtained by UAVs present higher resolution, which enables the capture of more subtle surface features. Nonetheless, it leads to an extremely limited proportion of small targets in the image. For instance, within the field of agricultural monitoring, small targets to be detected, such as small pest patches and water-deficient areas, may only occupy dozens or even several pixels. Additionally, the background of UAV remote sensing images typically contains various complex background information such as terrain, vegetation, and buildings, which will seriously interfere with the detection of small targets.

III. Small Target Detection Algorithm Based on Deep Learning

A. Principle of YOLO Series Algorithms

YOLO is a deep learning algorithm commonly used in the field of target detection. Taking YOLOv8 as an example, it regards target detection as a regression problem, adopting an end-to-end framework. In terms of the specific process, based on the input UAV remote sensing images, YOLOv8 employs convolutional and pooling layers to extract relevant features, thereby generating feature maps of different scales. On this basis, the anchor frame is preset on the feature map. By predicting the offset between the anchor frame and the real target, the category probability, and the confidence of the bounding box, YOLOv8 ultimately determines the position and category of the target.

As a whole, the network structure of YOLOv8 consists of an input end, backbone network, neck network, and output end. More specifically, first and foremost, the input end leverages Mosaic data enhancement technology to stitch images, thus enriching the training data. Secondly, utilizing CSPDarknet structure, the backbone network reduces the computation and enhances the feature propagation by cross-stage local network. Moreover, by fusing FPN and PAN structures, the neck network integrates features of diverse scales to improve the detection ability for small targets. Lastly, the output end outputs the bounding box and category information of the predicted target directly without the need for anchor boxes. However, YOLOv8 has not yet developed a specialized feature pyramid specifically designed for the small-target characteristics in UAV remote sensing images, leading to inadequate extraction of small-target features.

B. Principle of Faster R-CNN algorithm

Faster R-CNN is another classic target detection algorithm, which consists of the region proposal network (RPN) and the FAST R-CNN detection network. Among them, the main function of RPN is to generate a region of interest (RoI) that potentially contains the target. By sliding a small convolution kernel on the feature map obtained by the input image passing through the convolutional layer, it predicts the overlap between a plurality of anchor frames with different sizes and proportions and the real target for each sliding window, thereby screening out the anchor frames with higher foreground probability as RoIs. Furthermore, the Fast R-CNN detection network classifies the RoIs generated by RPN and regresses the bounding box, ultimately determining the category and precise position of the target. Its general framework is illustrated in Fig. 1.

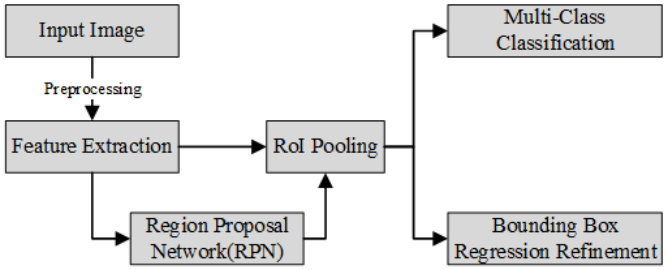


Fig. 1. Faster R-CNN general framework.

In Faster R-CNN, the loss function is denoted as:

$$L(\{p\},\{t_i\}) = \frac{1}{N_{cls}}\sum_i L_{cls}(p_i, p_i^*) + \lambda\frac{1}{N_{reg}}\sum_i p_i^* L_{reg}(t_i, t_i^*) \quad (1)$$

Where i denotes the index of the anchor box; p_i represents the probability that the anchor box contains an object; t_i denotes the predicted bounding box parameters; t_i^*represents the ground-truth bounding box parameters; p_i^* is used to determine whether there is an object in the anchor box, with 1 indicating presence and 0 indicating absence; N_{cls} and N_{reg} denote the respective quantities of samples utilized for classification and regression; λ represents the weight used to balance the classification loss and the regression loss.

The loss function of RPN encompasses classification loss and bounding box regression loss:

$$L_{cls}(p_i, p_i^*) = -log[p_i^* p_i + (1 - p_i^*)(1 - p_i)] \quad (2)$$

$$L_{reg}(t_i, t_i^*) = smooth_{L_1}(t_i - t_i^*) \quad (3)$$

Faster R-CNN minimizes computational overhead by sharing the backbone network between the RPN and Fast R-CNN, yet its two-stage detection pipeline still suffers from subpar real-time performance (typically 5-10 FPS) and exhibits limited feature extraction capability for small targets.

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Consequently, its accuracy in small target detection within UAV remote sensing images falls short of that achieved by YOLO series algorithms.

IV. Improved Architecture for UAV Remote Sensing Images

In object detection approaches, feature pyramids like FPN and PANet bolster multi-scale object recognition and localization abilities by employing efficient feature extraction and fusion techniques. Nevertheless, these network architectures, which generate high-level feature maps through successive downsampling and upsampling layers, are susceptible to losing features of small-scale objects. YOLOv8 integrates the strengths of both FPN and PANet to establish a more efficient multi-scale feature extraction framework. However, it fails to incorporate a feature pyramid structure that is better suited for small object detection, thereby posing challenges in accurately extracting feature information from small objects.

Building on this, this paper introduces a small object detection model featuring an enhanced feature pyramid, named FP-CAG-YOLO. The key enhancements encompass the design of a multi-scale feature fusion module, the integration of a compound attention mechanism (CAG) , and the refinement of the detection head structure, all aimed at mitigating the challenges of small object feature loss and inadequate background interference suppression in UAV remote sensing imagery. The overall architecture of the model is depicted in Fig. 2.

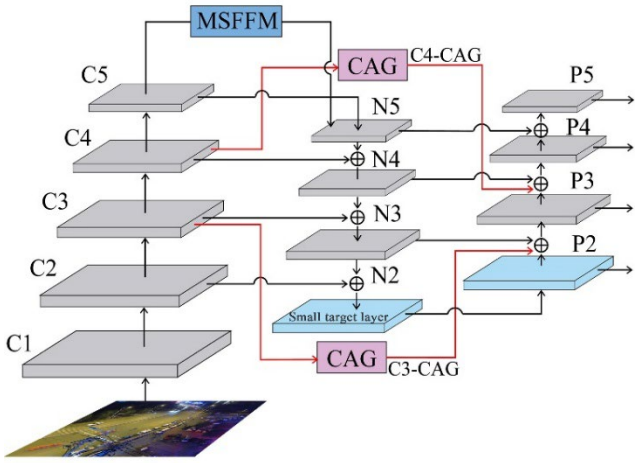


Fig. 2. Overall network structure of the improved model.

A. Multiscale Feature Fusion

Given the significant scale changes in small targets in UAV remote sensing images, this paper designs a multiscale feature fusion module, which obtains more abundant small target features through the fusion between different levels of feature maps of backbone network, as shown in Fig. 3. More specifically, through up-sampling and down-sampling operations, the proposed model adjusts the feature maps of different scales to the same resolution, thereby implementing element-by-element addition or splicing operations. As an example, by fusing the shallow feature map containing more detailed information with the deep feature map containing more semantic information through up-sampling, the proposed model endows the fused feature map with both detailed and semantic features, which is helpful to improve the accuracy of small target detection [9]. Assuming that the shallow feature map (C2) is represented as $F_{shallow}$, the mid-level feature map (C3) is represented as F_{mid} and the deep feature map (C4) is represented as F_{deep}, the fused feature map F_{fusion} with resolution adjusted by up-sampling and down-sampling can be calculated by the following equation, taking element-by-element addition as an example:

$$F_{fusion} = F'_{shallow} + F'_{mid} + F'_{deep} \tag{4}$$

Where $F'_{shallow}$, F'_{mid} and F'_{deep} respectively denote the shallow, mid-level and deep feature maps after resolution adjustment.

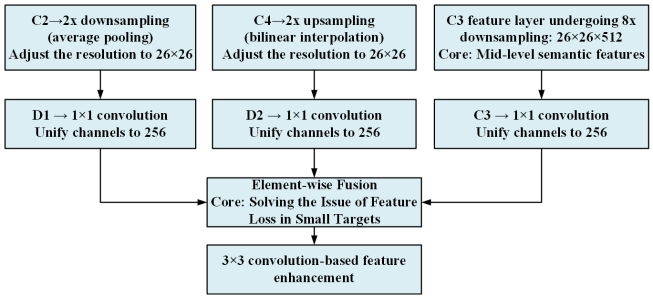


Fig. 3. Diagram of the multi-scale feature fusion of MSFFM module.

B. Attention Mechanism Embedding

To highlight the features of small targets in complex background, this paper embeds attention mechanism within the networ. As depicted in the Attention section of Fig. 2, the Attention module employs the Channel-Aware Fusion Attention Guided (CAG), as shown in Fig. 4. The channel attention mechanism CAG comprises two branches. The first branch employs maximum pooling to process the input feature maps, while the second branch utilizes average pooling for separate processing and analysis of the feature information. Subsequently, the data flows into a fully connected layer, and the desired attention weights are ultimately derived through element-wise summation and the Sigmoid activation function. CAG operates on the final output feature map, which aids in addressing information confusion arising from the feature pyramid feature fusion process, which can easily lead to false positives and false negatives. It mitigates the influence of complex backgrounds and conflicts on detection, thereby enabling the model to focus more on the desired components. This significantly enhances the detection capability for small remote sensing targets. To further improve the suitability of the cross-level reinforced feature pyramid network for small object detection tasks in aerial images, we have removed the detection head corresponding to C5 at the 32-fold downsampling point in the original PANet structure and added a "small object layer" along with its corresponding detection head (P2) at the 4-fold downsampling point C2. This modification emphasizes the advantage of detailed information for small objects and enhances the detection accuracy of such objects.

The formula for the CAG attention mechanism can be expressed as:

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$$CAG(X) = \sigma\left(fc_1\left(Avgpool(x)\right) + fc_2\left(Maxpool(x)\right)\right) \quad (5)$$

$$R_i = \mathrm{CAG}(I) \odot P_i \quad (6)$$

Where $CAG(X)$represent the channel attention function; σ is represented as the Sigmoid function; fc refers to fully connected operation; i represents the number of layers in the feature pyramid; P_i represents the output result of the feature pyramid; R_i represents the final output result after attention mechanism.

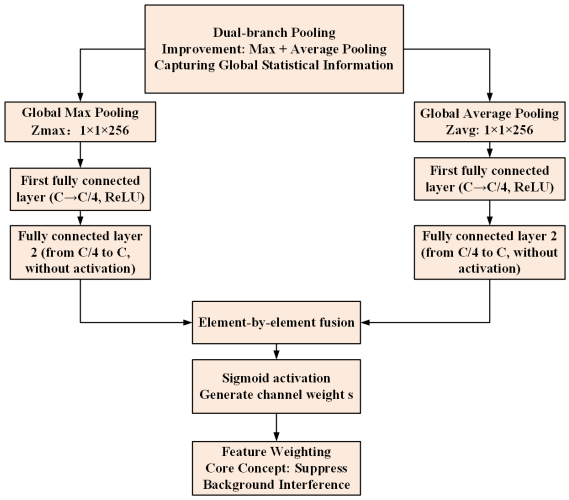


Fig. 4. Diagram of the Channel-Aware Fusion Attention Guided of CAG module.

C. Structural optimization of the detection head

To further improve the model's capability in detecting small targets, this paper optimizes the detection head structure of YOLOv8 by removing the C5 detection head at the 32x downsampling stage in PANet. The C5 feature map, which undergoes 32x downsampling (corresponding to 10×10 pixels based on a 640×640 input image), is primarily utilized for detecting large targets and offers minimal contribution to small target detection. Eliminating this detection head can reduce computational overhead by 15%-20% and enhance the model's real-time performance. Additionally, a new C2 "small target layer" detection head (P2) is introduced at the 4x downsampling stage. The C2 feature map, downsampled by 4 times (corresponding to 160×160 pixels), contains the most abundant detail information. The newly added P2 detection head is specifically designed to detect small targets (ranging from 5 to 30 pixels), thereby reinforcing the supportive role of detailed information in small target detection, as shown in Fig. 5.

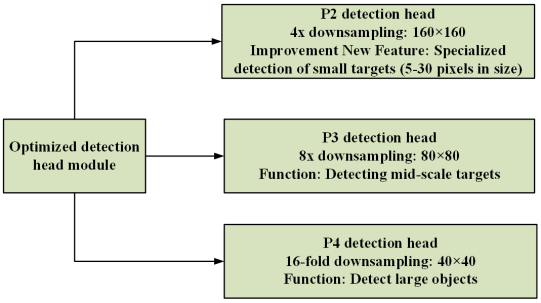


Fig. 5. Diagram of the optimized detection head module.

The optimized detection head includes three scales: P2 (with 4x downsampling, at a resolution of 160×160), P3 (with 8x downsampling, at a resolution of 80×80), and P4 (with 16x downsampling, at a resolution of 40×40), which correspond to, medium, and large targets, respectively. Specifically, the P2 detection head is tailored for small target detection, improving the accuracy of small target detection by predicting their bounding boxes, class probabilities, and confidence scores.

V. Experiment and Result Analysis

A. Experimental Dataset

This research employs the publicly accessible VisDrone dataset, which is UAV dataset, urban block scenes covering day and night. Particularly, a series of small targets including pedestrian, bicycle, car, van, truck, tricycle, bus and motor are marked, as shown in Fig. 6. The dataset encompasses 6,471 images in total, which are divided into training, verification, and testing sets according to the ratio of 7:2:1.



Fig. 6. Partial samples from the datasets.

B. Experimental Environment and Parameter Setting

The experiment leverages a computer equipped with NVIDIA RTX 3090 GPU, adopting PyTorch framework. This model employs an initial learning rate of 0.001 and SGD optimizer with momentum of 0.9 and weight attenuation of 0.0005. Under the training condition of batch size of 16, a total of 300 epoch were trained.

C. Evaluation Indicators

Three indicators, encompassing mean average precision (mAP), recall rate, and frames per second (FPS), are utilized to evaluate the proposed model. Specifically, by comprehensively taking the accuracy under different recall rates into account, mAP comprehensively reflects the detection performance. The recall rate reflects the proportion of real targets detected. FPS measures the number of image frames processed by the model per second, thereby reflecting the real-time performance.

D. Comparative Experiment on Target Detection

Experimental results show that the improved YOLO model achieved excellent performance in the test set. As compared with the original YOLOv8 and Faster R-CNN, demonstrating its

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exceptional capability in detecting targets amidst complex backgrounds and across various scales.

The FPE-CAG-YOLO model exhibits outstanding performance on the VisDrone. In comparison to the traditional YOLOv8 and Faster R-CNN models, the improved model utilizes Cross-level feature pyramid network as its feature extraction network and integrates the multi-scale feature fusion architecture, demonstrating its superiority in target detection. Table 1 presents the mAP values and FPS of various networks on the VisDrone dataset. As shown in Fig. 7, the performance variation trends of these three different models are presented.

TABLE I. COMPARATIVE EXPERIMENT ON VISDRONE

Item Model	Recall	mAP50	mAP50:95	FPS
Faster R-CNN	41.81	42.68	25.9	10
YOLOv8s	46.62	48.65	30.10	57
FPE-CAG-YOLO	54.44	57.25	38.68	48

The results show that the improved model achieves a mAP50 of 57.25% on the VisDrone dataset. Concretely, compared to the baseline model, the improved model demonstrates a 8.6% improvement in mAP50.

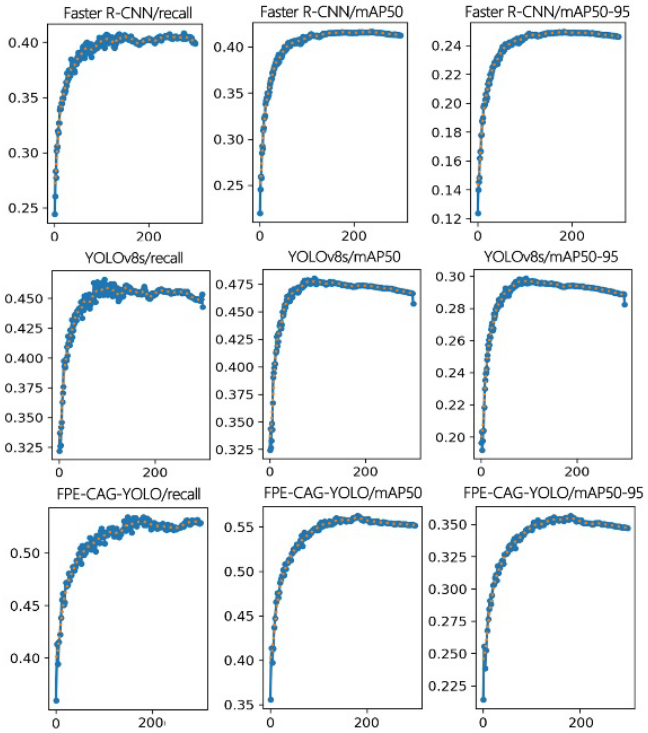


Fig. 7. Detection results.

E. Ablation Experiment

To validate the effectiveness of each improved module in the FPE-CAG-YOLO model, an ablation study was conducted. Using YOLOv8 as the baseline model, we sequentially incorporated the multi-scale feature fusion module (MSMFF), composite attention mechanism (CAG), and head optimization (HO), and then integrated all modules to form the FPE-CAG-YOLO model. The performance of each model variant was evaluated on the VisDrone test set, with the results presented in Table 2.

TABLE II. ABLATION EXPERIMENT ON VISDRONE

Item Model	MSMFF	CAG	HO	Recall	mAp 50	mAP5 0:95	FPS
YOLOv8s	×	×	×	46.62	48.65	30.10	57
YOLOv8s+ MSMFF	√	×	×	49.12	53.47	33.02	48
YOLOv8s+ CAG	×	√	×	50.32	52.96	32.17	52
YOLOv8s+ HO	×	×	√	51.86	53.28	33.87	55
FPE-CAG-YOLO (MSMFF+ CAG+HO)	√	√	√	54.44	57.25	38.68	45

From Table 2, we can draw the following conclusions:

1) The function of the Multi-scale Feature Fusion (MSMFF) module: Upon incorporating MSMFF, the model's mAP50 improved by 4.82%, and the recall rate rose by 2.5%. This demonstrates that MSMFF can effectively integrate shallow detail features with deep semantic features, minimize the loss of small target features, and enhance detection accuracy. The FPS dropped from 57 to 48, primarily attributable to the heightened computational demands of feature adjustment and convolution operations stemming from multi-scale feature fusion.

2) The function of the Composite Attention Mechanism (CAG): Upon integrating CAG, the model's mAP50 improved by 4.31%, and the recall rate rose by 3.7%. This demonstrates that CAG can effectively mitigate interference from complex backgrounds, amplify small target features, and minimize false detections and missed detections. The FPS dropped from 57 to 50, with only a slight increase in computational load, since CAG primarily relies on pooling and fully connected layers, resulting in a relatively small number of parameters (approximately 2%-3% of the total model parameters).

3) The function of Head Optimization (HO): After incorporating HO, the model's mAP50 improved by 4.63%, and the recall rate increased by 5.24%. This demonstrates that eliminating the C5 detection head reduces redundant computations, while introducing the C2 detection head enhances the exploitation of detailed information from small targets, significantly boosting the recall rate for small target detection. The FPS dropped from 57 to 55, yet the reduction in computational load is negligible, as the computational savings from removing the C5 detection head are largely counterbalanced by the additional computations introduced by the C2 detection head.

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4) The synergistic effect among modules: Upon integrating *MS*MFF, CAG, and HO, the model witnessed an 8.6% increase in mAP50 and a 7.82% rise in recall rate, demonstrating a synergistic interaction among the modules. Specifically, *MS*MFF supplies comprehensive features for small objects, CAG strengthens these features while mitigating interference, and HO refines the detection head to better suit small objects. The synergy of these three modules leads to a marked enhancement in detection performance. Although the FPS decreases to 45, it still fulfills real-time processing demands, thereby validating the rationality of each module's design.

F. Visualized results of object detection

To effectively demonstrate the detection performance of the improved model, we selected test images from the VisDrone dataset in four different scenarios for detection. Fig. 8 presents the detection results of the improved Faster R-CNN model in different scenarios, including complex backgrounds, dense small targets, and poor imaging quality.



Fig. 8. Detection performance.

VI. CONCLUSIONS

To sum up, this paper investigates the real-time small target detection method by UAV remote sensing image based on deep learning, analyzing the principle of existing classical algorithms in detail. Simultaneously, this paper innovatively proposes an improved architecture based on the characteristics of UAV remote sensing images. Through rigorous experimental validation, the improved model achieves satisfactory results in both the accuracy and real-time performance of small target detection, thus furnishing a practical and effective solution for the practical application of UAV remote sensing images in small target detection. Focusing on exploring more efficient network structures and algorithms, the follow-up research will not only further improve the model’s ability to detect complex scenes and small targets but also reduce the calculation cost to promote the wide application of UAV remote sensing in diverse fields.

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